

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Full-stack engineer with production experience building React + TypeScript applications, backend APIs, and AI-native workflows. At Isomer AI, shipped 10+ features across 7 deployments on a TypeScript/Node.js/React/PostgreSQL stack - including real-time event pipelines, role-based access control, and document processing systems. Built AlgoPulse as a React + Node.js full-stack platform capable of handling 2,000 async AI inference requests/second. Excited by ambitious, zero-to-one product challenges like rebuilding a core dashboard from scratch with modern tooling and embedded AI.

CORE COMPETENCIES

React & TypeScript

Next.js

Full-Stack Product Engineering

Node.js / Express APIs

Real-Time Streaming (Pusher/WebSockets)

PostgreSQL & Redis

AI-Native Development

Multi-Model LLM Integration

Docker & CI/CD

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - San Francisco, CA (Remote)

- Shipped **10+ production features** on a TypeScript/Node.js/React/PostgreSQL stack across 7 deployments - backend APIs, UI components, and real-time Pusher event pipelines from design to production
- Built **role-based access control** (RBAC) system, document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), and multi-model AI agent tooling across Claude, GPT-4o, and Gemini using MCP
- Integrated Auth0 authentication, Redis caching, and Sentry observability; **contributed 394 commits** to the production codebase across a complex microservices architecture

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Built a cloud-based genomic data platform with **React + JBrowse 2**, enabling interactive data exploration for 100+ researchers across 500GB of datasets deployed on Google Cloud Run
- Applied hyperparameter tuning and time-series ML analysis, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%
- Automated deployments on AWS & GCP with Grafana monitoring; supported model validation and red-teaming

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a data-driven support ticket system; implemented **Kafka streaming + Spark batch pipelines** reducing latency by 20% and processing time by 15%

PROJECTS

AlgoPulse - Full-Stack AI Platform Independent · 2025

Built a **React + Node.js full-stack platform** for AI-powered algorithmic analysis. Designed responsive UI with real-time streaming of AI responses from server-sent Supabase Edge Functions directly to the client. Scaled to **2,000 async AI inference requests/second** with Temporal.io workflow orchestration and Kafka for state synchronization - zero state loss under load.

TypeScript · React · Node.js · Supabase Edge Functions · Temporal.io · Kafka · PostgreSQL

Glance AI - Privacy-First RAG Dashboard Independent · 2025

Built a compliance-ready RAG system with per-tenant data isolation at the vector retrieval layer for fintech/healthcare. Implemented deterministic output architecture ensuring factual correctness for regulated environments.

Python · LLMs · Vector Databases · RAG

PerCent - GenAI Finance App HackNC24 Top 10 of 100+ teams

Full-stack GenAI finance app with goal tracking, expense logging, and an **AI chatbot (RAG + Qwen-14B-Chat)** that reduced manual financial research by 30%. Built complete MVP in 48 hours.

React · Node.js · Hugging Face · Qwen-14B-Chat · Selenium

Sanjeevani - Health App Top 1% of 1,100+ teams

Full-stack React application with ML-powered predictive health insights achieving 87% accuracy using real-time sensor data. Shortlisted top 12 of 1,100+ teams for UI innovation.

React · Node.js · Python · ML classification models

Mapify - ML & NLP Tool IEEE Published

Built T5-transformer pipeline converting 1,500+ PDFs/images into mind maps and flowcharts; reduced academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python · T5 Transformer · NLTK · NetworkX

EDUCATION

North Carolina State University - MS Computer Science Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms

Mumbai University - BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

- Frontend:** React, TypeScript, Next.js, JavaScript, Tailwind CSS, Angular
- Backend:** Node.js, Express, PostgreSQL, Redis, Prisma, REST APIs, GraphQL
- AI/ML:** OpenAI API, Anthropic Claude, Google Gemini, LangChain, MCP, RAG, Hugging Face, TensorFlow
- Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, CI/CD, Terraform, Grafana, Temporal.io, Kafka
- Languages:** Python, Java, SQL, NoSQL, C, R, Ruby